ECE1508 Group Project Proposal

Residual Deep RL for Quadrotor Flight Under Downwash and Its Generalization to Unseen Aerodynamic Effects

Yi Lok
Farnaz Moayyedian
Chen Jiarong Edwin

October 2026

1 Main Idea and RL Component

When a quadrotor flies beneath another, the upper vehicle's downwash causes thrust loss and attitude disturbances that a standard PID controller compensates for poorly. Inspired by ProxFly [1], we train a residual policy with Proximal Policy Optimization (PPO) [2] that adds small corrections to a PID controller, entirely in simulation. In our main scenario, the controlled drone hovers or tracks a circle while a second, scripted drone passes overhead at varying vertical separations. We then test the same policy, without retraining, on near-ground hovering, where ground effect pushes the drone up instead of down.

We address two research questions:

1. How much does residual learning improve tracking accuracy and training efficiency compared with the PID alone and with end-to-end PPO?
2. Does a policy trained only under overhead downwash generalize to ground effect, a disturbance it never experienced?

2 MDP Formulation

- **Agent:** the residual policy; the actor and critic are neural networks trained with PPO.
- **Environment:** the quadrotor physics, the PID controller, the scripted upper drone, and external disturbances.
- **Observation (20 values):** errors in position, velocity, attitude and angular rate relative to the reference (12 values), the PID's motor commands (4 values), and the previous residual action (4 values).
- **Action (4 values):** a bounded correction added to each of the PID's four motor commands.
- **Reward:** penalties on position error, attitude error, and command magnitude and oscillation, plus a small per-step survival bonus.
- **Episodes:** fixed duration, terminated early on a crash. Mass, propeller thrust efficiency, and random external forces and torques are randomized each episode. Since these are not observed by the policy, the problem is partially observable.

Baselines. PID only; end-to-end PPO without the PID; and, for the ground-effect test, an oracle residual policy trained directly with ground effect, as an upper bound.

3 Key Outputs

- A public GitHub repository with the two-drone environment, disturbance randomization, training and evaluation scripts, and a demo notebook with flight videos.
- Results averaged over 5 random seeds: position and attitude RMSE, maximum altitude drop during the pass, crash rate, and learning curves, for overhead passes at 0.25, 0.5 and 0.75 m separation and near-ground hovering at 5 and 10 cm.
- Plots of the residual commands showing when and how the policy compensates for each disturbance.
- The final report and presentation.

4 Requirements and Plan

Software: Python, PyTorch, Stable-Baselines3 [3], and gym-pybullet-drones [4], which supports multiple drones and includes downwash and ground-effect models. ProxFly's public code serves as a reference.

Compute: The simulator runs on CPU, so laptops and Google Colab are sufficient; a departmental lab machine, if available, will run additional seeds.

Data: None required, since all experience is generated in simulation.

Preparation Plan:

- Weeks 1–2: environment setup and PID baseline.
- Weeks 3–5: residual PPO for hovering under overhead passes.
- Weeks 6–7: circle tracking and the end-to-end baseline.
- Weeks 8–9: ground-effect generalization tests and the oracle.
- Week 10: report and presentation.

References

[1] R. Zhang, D. Zhang, and M. W. Mueller, "ProxFly: Robust control for close proximity quadcopter flight via residual reinforcement learning," arXiv:2409.13193, 2024.

[2] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[3] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.

[4] J. Panerati, H. Zheng, S. Zhou, J. Xu, A. Prorok, and A. P. Schoellig, "Learning to fly—a gym environment with PyBullet physics for reinforcement learning of multi-agent quadcopter control," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2021, pp. 7512–7519.